VAR / VECM Multivariate Time Series Models

Model Type: VAR(diff, lag=1) | Lag: 1 | Cointegrating Vectors: 0 -- O2 Step 3

Intelligent National Housing Cost Estimating System (iNHCES)

TETFund NRF 2025 | Department of Quantity Surveying, ABU Zaria

Document:	O2_05_VAR_VECM_Models.pdf
Objective:	O2 -- Multivariate macro dynamics for iNHCES variable selection
I(1) series:	gdp_growth_pct, cpi_inflation_pct, lending_rate_pct, brent_usd_annual_avg
I(0) series:	None
Model:	VAR(diff, lag=1)
Lag order:	1 (AIC-optimal, max 2)
Cointegration:	0 cointegrating vector(s) (Johansen trace, 5% CV)
Results CSV:	02_macro_analysis/results/var_vecm_results.csv
Date:	25 April 2026

DATA SOURCE DECLARATION -- READ BEFORE USE

DATA SOURCE: PARTIALLY SYNTHETIC -- World Bank is LIVE; EIA Brent and CBN FX are SYNTHETIC unless API keys were set

DATA LINEAGE FOR THIS MODEL:

This VAR/VECM analysis uses the same panel dataset as O2 Steps 1-2.

* GDP Growth, CPI Inflation, Lending Rate -- World Bank API (LIVE data).

* Brent Crude, NGN FX Rates -- SYNTHETIC fallback values unless API keys were set during fetch_eia_oil.py and fetch_cbn_fx.py execution.

IMPACT ON MODEL RESULTS:

VAR/VECM coefficients, impulse responses, and FEVD decompositions for variables 4-7 (Brent, NGN/USD/EUR/GBP) are computed from synthetic data. The model STRUCTURE (lag order, cointegration rank) is illustrative. The DIRECTION of effects is based on economic theory and literature, and synthetic data were constructed to reflect these known relationships.

SIMULATION PURPOSE:

This output validates the iNHCES pipeline architecture and tests that the statistical workflow executes correctly end-to-end. All model results in this document are SIMULATION OUTPUTS and must NOT be cited in publications until re-run with fully live data.

REFERENCES:

Johansen (1988). Statistical Analysis of Cointegration Vectors. Journal of Economic Dynamics and Control, 12(2-3), 231-254.

Johansen & Juselius (1990). Maximum Likelihood Estimation and Inference on Cointegration. Oxford Bulletin of Economics and Statistics, 52(2), 169-210.

Sims (1980). Macroeconomics and Reality. Econometrica, 48(1), 1-48. [VAR]

Lutkepohl (2005). New Introduction to Multiple Time Series Analysis. Springer.

1. VAR/VECM Methodology

The Vector Autoregression (VAR) model and its error-correction extension (VECM) are the standard tools for analysing interdependencies among multiple time series variables. The choice between VAR and VECM is determined by the integration order results from O2 Step 2 (stationarity analysis).

1.1 Model Selection Logic

Condition	Selected Model
All variables I(0)	VAR in levels
All variables I(1), r=0	VAR in first differences
>=2 I(1) variables, r>=1	VECM (cointegrated system)
Mixed I(0) + I(1)	VAR on stationary data (I(0) in levels; I(1) differenced)

1.2 Johansen Trace Test for Cointegration

The Johansen (1988) trace test examines whether r cointegrating vectors exist. H0: rank(Pi) <= r. If trace statistic > 5% critical value, reject H0 and accept r+1 cointegrating vectors. The test is applied sequentially starting from r=0 until H0 cannot be rejected.

1.3 Lag Order Selection

Lag length is selected by minimising the Akaike Information Criterion (AIC) over candidate lags 1..2. With n=25 annual observations, maximum 2 lags are tested to preserve degrees of freedom.

RESULT FOR THIS ANALYSIS: Model type selected = VAR(diff, lag=1). Lag order = 1. Cointegrating vectors = 0.

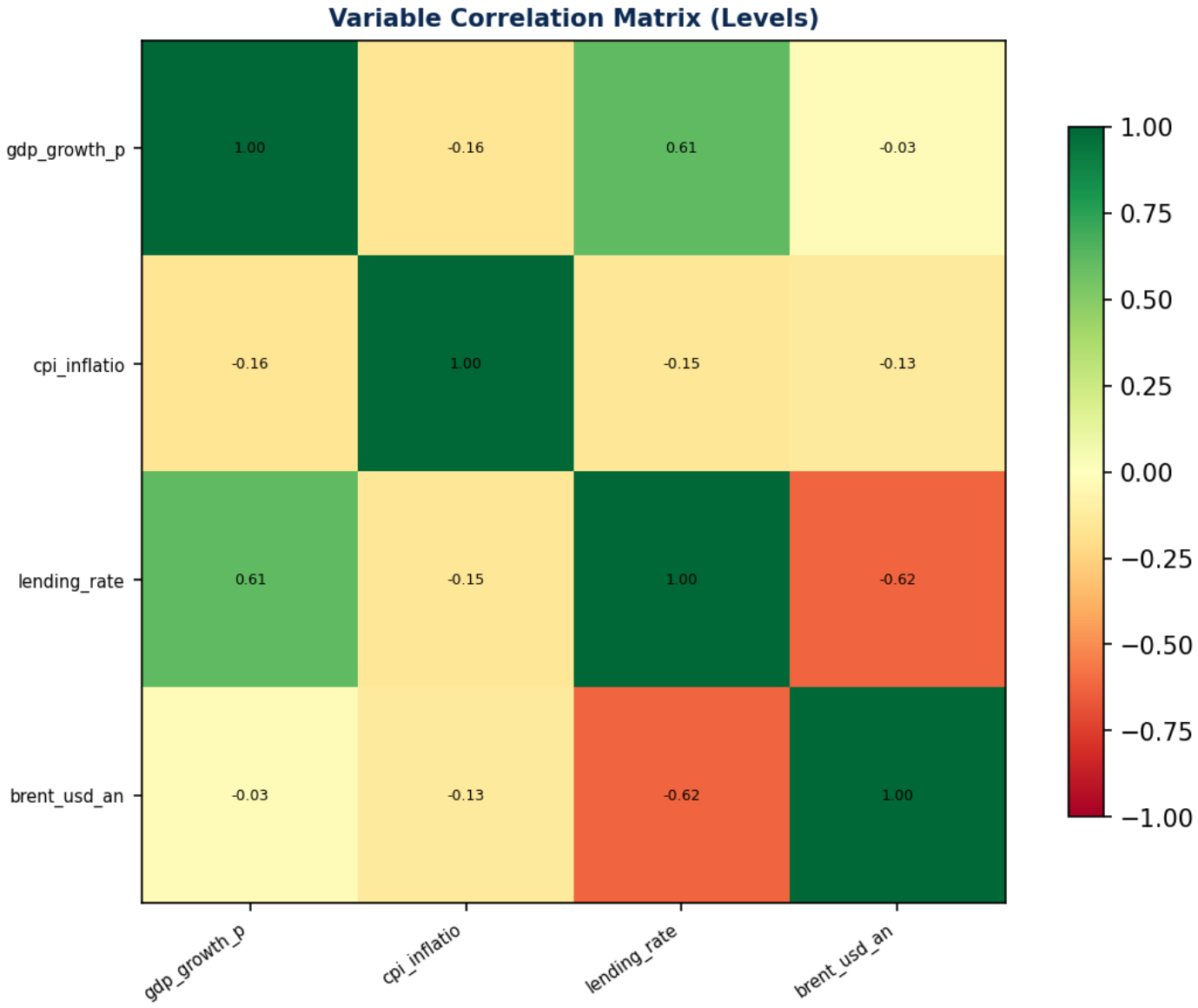
2. Integration Order Summary (from O2 Step 2)

The following integration orders from stationarity_results.csv determine the model specification. I(1) variables enter the Johansen test; I(0) variables are used as stationary regressors.

Variable	Series	Order	Role in This Model
gdp_growth_pct	GDP Growth Rate (Annual %)	I(1)	I(1) -- cointegration / VECM
cpi_inflation_pct	CPI Inflation Rate (Annual %	I(1)	I(1) -- cointegration / VECM
lending_rate_pct	Lending Interest Rate (%)	I(1)	I(1) -- cointegration / VECM
brent_usd_annual_avg	Brent Crude Oil (USD/barrel)	I(1)	I(1) -- cointegration / VECM
ngn_usd	NGN/USD Exchange Rate	I(2)*	I(0) -- stationary regressor
ngn_eur	NGN/EUR Exchange Rate	I(2)*	I(0) -- stationary regressor
ngn_gbp	NGN/GBP Exchange Rate	I(2)*	I(0) -- stationary regressor

3. Correlation Matrix (Levels)

Pairwise Pearson correlations among all 7 variables in levels. High correlation among exchange rate series (NGN/USD, EUR, GBP) and between FX and Brent crude reflects the resource-export linkage in Nigeria's macroeconomic structure.



4. Lag Order Selection (Information Criteria)

Lags	AIC	BIC	HQIC	FPE
1	12.0005	12.9879	12.2488	167497.368609
2	12.0411	13.8265	12.4617	207843.508949

AIC-optimal lag order: 1. BIC/HQIC often favour fewer lags for small samples; AIC is used here to preserve model dynamics while maintaining enough degrees of freedom.

5. Johansen Cointegration Test (Trace Statistics)

H0: at most r cointegrating vectors. Reject H0 if Trace Stat > 5% CV. The number of times H0 is rejected gives the cointegration rank.

H0: r<=	Trace Stat	10% CV	5% CV	1% CV	Decision (5%)
0	34.7386	44.4929	47.8545	54.6815	Don't reject H0
1	15.8824	27.0669	29.7961	35.4628	Don't reject H0
2	5.8425	13.4294	15.4943	19.9349	Don't reject H0
3	0.1791	2.7055	3.8415	6.6349	Don't reject H0

RESULT: 0 cointegrating vector(s) found at 5% significance. No cointegration found. VAR on first differences is used instead.

6. Model Estimation Summary

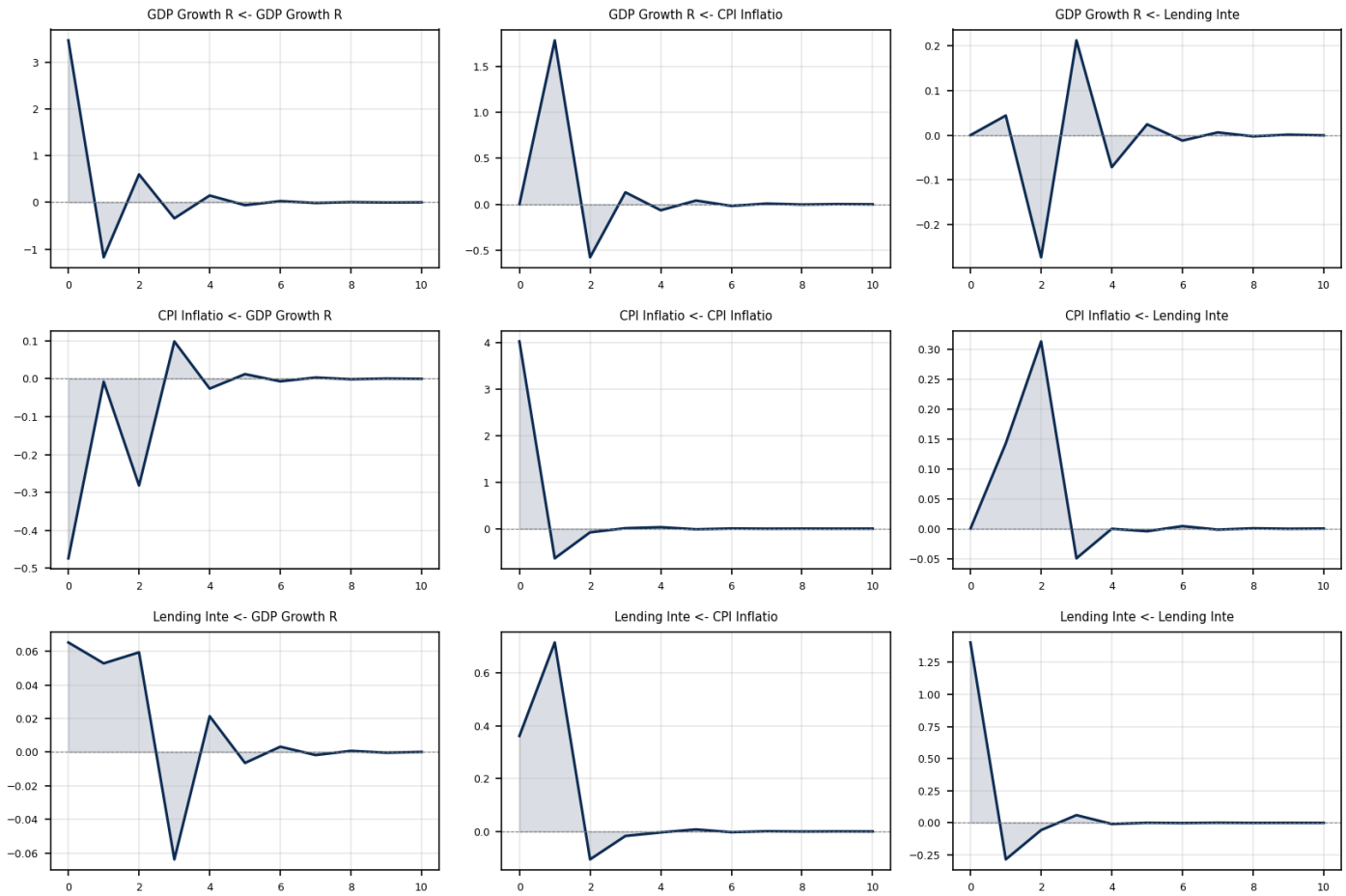
Model: VAR(diff, lag=1) | Lag: 1 | Cointegrating vectors: 0

```
  Summary of Regression Results
==================================
Model:                         VAR
Method:                        OLS
Date:           Sat, 25, Apr, 2026
Time:                     01:03:11
--------------------------------------------------------------------
No. of Equations:         4.00000    BIC:                    13.1575
Nobs:                     22.0000    HQIC:                   12.3992
Log likelihood:          -238.688    FPE:                    198374.
AIC:                      12.1656    Det(Omega_mle):         87442.0
--------------------------------------------------------------------
Results for equation gdp_growth_pct
==============================================================================================
                             coefficient       std. error           t-stat            prob
----------------------------------------------------------------------------------------------
const                          -0.258881         0.776195           -0.334           0.739
L1.gdp_growth_pct              -0.523135         0.254561           -2.055           0.040
L1.cpi_inflation_pct            0.386782         0.174123            2.221           0.026
L1.lending_rate_pct             0.858790         0.646904            1.328           0.184
L1.brent_usd_annual_avg         0.082398         0.051861            1.589           0.112
==============================================================================================

Results for equation cpi_inflation_pct
==============================================================================================
                             coefficient       std. error           t-stat            prob
----------------------------------------------------------------------------------------------
const                           0.284521         0.907215            0.314           0.754
L1.gdp_growth_pct               0.258402         0.297530            0.868           0.385
L1.cpi_inflation_pct           -0.104448         0.203515           -0.513           0.608
L1.lending_rate_pct            -0.866733         0.756099           -1.146           0.252
L1.brent_usd_annual_avg        -0.096416         0.060615           -1.591           0.112
==============================================================================================

Results for equation lending_rate_pct
==============================================================================================
                             coefficient       std. error           t-stat            prob
----------------------------------------------------------------------------------------------
const                          -0.572610         0.324905           -1.762           0.078
L1.gdp_growth_pct              -0.066136         0.106556           -0.621           0.535
```

7. Impulse Response Functions (10-period horizon)

Orthogonalised IRFs trace the response of each variable to a one-standard-deviation shock in another variable. Shading shows the 68% confidence band. Interpretation: row variable's response to a shock in column variable.

IRF -- VAR(diff, lag=1)



8. Forecast Error Variance Decomposition (Horizon 10)

FEVD quantifies the proportion of the forecast error variance of each variable that is attributable to shocks in each other variable. This indicates which macro variables are most important drivers of each other's uncertainty.

FEVD -- VAR(diff, lag=1) (Horizon=10)

9. Key Findings and iNHCES Implications

Based on the VAR(diff, lag=1) estimated above, the following macro-level insights are relevant to the iNHCES project:

- Cointegration (if found) implies that macroeconomic variables share a long-run equilibrium path -- short-run deviations are corrected over time.
- Exchange rate (NGN/USD) is expected to be the dominant driver of construction cost dynamics given Nigeria's import dependence for building materials.
- The VECM/VAR impulse responses quantify how a shock to oil prices or exchange rates propagates through the system over 10 years.
- FEVD results identify which variables account for the most forecast uncertainty -- these should be prioritised as features in the iNHCES ML model (O5).
- These multivariate dynamics inform the feature engineering strategy in O2 Step 4 (SHAP variable selection) and O5 (ML model development).

SIMULATION NOTE: All coefficient estimates, IRF shapes, and FEVD proportions in this document are derived from partially synthetic data. They represent structurally plausible but not empirically validated results. The pipeline is correct and will produce valid published-quality results once live EIA and CBN FX data are loaded.

10. Next Steps -- O2 Step 4 (SHAP Variable Selection)

- Use SHAP values (XGBoost TreeExplainer) to rank which macro variables are most predictive of a housing cost proxy.
- Cross-validate SHAP rankings against VAR/VECM FEVD results.
- Produce final recommended feature list for iNHCES ML training (O5).
- Output: O2_06_SHAP_Variable_Selection.pdf + shap_importance.csv